Example

```
>>> Even_List = list(range(2,11,2))
>>> print(Even_List)
[2, 4, 6, 8, 10]
```

In the above code, **list()** function takes the result of **range()** as Even_List elements. Thus, Even_List list has the elements of first five even numbers.

Similarly, we can create any series of values using **range()** function. The following example explains how to create a list with squares of first 10 natural numbers.

Example : Generating squares of first 10 natural numbers

```
squares = [ ]
for x in range(1,11):
        s = x ** 2
        squares.append(s)
print (squares)
```

In the above program, an empty list is created named "squares". Then, the for loop generates natural numbers from 1 to 10 using **range()** function. Inside the loop, the current value of x is raised to the power 2 and stored in the variables. Each new value of square is appended to the list "squares". Finally, the program shows the following values as output.

Output

[1, 4, 9, 16, 25, 36, 49, 64, 81, 100]

9.1.10 List comprehensions

List comprehension is a simplest way of creating sequence of elements that satisfy a certain condition.

Syntax:

List = [expression for variable in range]

Example : Generating squares of first 10 natural numbers using the concept of List comprehension

```
>>> squares = [ x ** 2 for x in range(1,11) ]
>>> print (squares)
```

Output:

[1, 4, 9, 16, 25, 36, 49, 64, 81, 100]

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